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Expert-Annotated Optical Microscopy Images of Human Sperm for Detection and DNA Fragmentation Assessment

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We present a collection of expert-labeled image datasets designed to support automated analysis of human sperm in the context of male fertility diagnostics. The collection includes three datasets: a raw set of over 400 high-resolution, stained bright-field images containing multiple sperm cells; a binary classification set with cropped image patches labeled as either sperm or non-sperm; and a multiclass dataset of individual sperm cells categorized by DNA fragmentation level, based on halo size (large, medium, small, or no halo). All annotations were independently performed by five experienced embryologists, with final labels determined by majority agreement. Together, these datasets provide a foundation for training and benchmarking machine learning models for sperm detection, classification, and chromatin integrity assessment. The data are openly available to facilitate reproducibility and support further research in reproductive health.

Background & Summary

Infertility is a widespread issue affecting 8–12% of couples worldwide, with male factors contributing to approximately 40–50% of these cases^{1,2}. Male infertility is mostly related to various factors, including sperm quality, motility, and DNA integrity³. While semen analysis has long been the standard diagnostic test for male infertility, many cases remain unexplained, highlighting the need for more advanced diagnostic methods. Among these tools, sperm DNA fragmentation testing has emerged as a key determinant of male fertility, as high levels of DNA fragmentation are associated with reduced fertilization success, increased miscarriage rates, and lower live birth rates^{4–6}.

Several assays are currently available for assessing sperm DNA fragmentation (SDF), each with specific strengths and limitations. The TUNEL (Terminal deoxynucleotidyl transferase dUTP Nick End Labeling) assay is considered a gold standard because it directly detects DNA strand breaks; however, it requires fluorescence microscopy or flow cytometry, is time-consuming, and shows inter-laboratory variability^{7,8}. The Comet assay (single-cell gel electrophoresis) provides quantitative assessment of DNA damage at the single-cell level, but its results are highly operator-dependent and lack standardized protocols⁹. The Sperm Chromatin Structure Assay (SCSA) measures the susceptibility of sperm DNA to acid-induced denaturation using acridine orange staining, and it has demonstrated strong clinical utility with high reproducibility and suitability for large sample sizes. Nevertheless, it requires expensive flow cytometry and technical expertise⁵.

The Sperm Chromatin Dispersion (SCD) test, in contrast, is relatively simple, cost-effective, and does not require advanced instrumentation. It distinguishes sperm with intact DNA (displaying a large dispersion halo) from those with fragmented DNA (without a halo). Clinical investigations have shown that SCD correlates with reproductive outcomes; however, its sensitivity is lower than that of TUNEL and Comet assays, and the interpretation may be subjective. Despite these limitations, SCD has become widely used in clinical practice because it provides a practical balance between accessibility, diagnostic value, and applicability in fertility clinics^{10,11}.

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As a result, sperm analysis plays a critical role in fertility assessments, helping clinicians evaluate the reproductive potential of male patients and determine appropriate interventions^{12,13}.

Traditional methods for sperm analysis, particularly sperm morphology evaluation and DNA fragmentation testing, are labor-intensive and subjective. Methods like the sperm chromatin dispersion (SCD) test rely on manual observation of sperm halos under a microscope to assess DNA fragmentation^{14,15}. However, these assessments are sometimes prone to intra- and inter-observer variability, which can lead to inconsistent results. Also, manual evaluation is limited in throughput, restricting the number of sperm cells that can be analyzed in a given sample. These challenges showcase the need for automated, objective methods that can provide more reliable and high-throughput assessments of sperm quality, particularly in clinical settings where accuracy is critical for decision-making.

Recent advances have increasingly applied machine learning to automate sperm analysis tasks such as detection, segmentation, morphology classification, motility assessment, and DNA fragmentation evaluation¹⁶. demonstrated that stain-free interferometric phase microscopy combined with ML can classify individual sperm cells objectively, while¹⁷ integrated microfluidic platforms with computational analysis to improve sperm quality evaluation in high-throughput settings.

Deep learning frameworks have also been developed for video-based analysis, such as the RoSTFine model by¹⁸, which simultaneously assesses sperm morphology and motility, and earlier work by¹⁹, who applied convolutional neural networks to predict sperm motility directly from semen videos. In addition, dedicated architectures such as TOD-CNN have achieved high performance in sperm detection from microscopy images, supporting automated quality control²⁰. Recent contributions in this field also include the work of²¹, who introduced a modified U-Net model for sperm segmentation and DNA fragmentation analysis, demonstrating the ability of deep learning methods to capture subtle structural differences in sperm cells. Similar ML-based approaches have also been successfully extended to other areas of reproductive medicine^{22,23}. Collectively, these studies demonstrate that machine learning has already begun to transform reproductive health by enabling objective, reproducible, and scalable image-based diagnostics, reinforcing the value of publicly available sperm datasets for advancing this field²⁴.

In this context, we introduce a collection of datasets aimed at enabling automated sperm detection and DNA fragmentation analysis through image-based machine learning.

To support this goal, we assembled a collection of three datasets centered on stained, high-resolution images of human sperm samples prepared using the SCD method. The first dataset comprises over 400 full-field bright-field images, each capturing multiple sperm cells with varying morphologies and halo characteristics. These images reflect the natural heterogeneity seen in clinical samples and serve as a foundation for image segmentation, unsupervised analysis, and model development.

In addition to the raw images, we constructed two labeled datasets derived from manually cropped sperm and non-sperm entities. The first labeled dataset is designed for binary classification, distinguishing between sperm cells and non-sperm structures such as debris or round cells. The second supports multiclass classification of individual sperm cells into four categories based on halo size (large, medium, small, or none) as a proxy for DNA fragmentation. All annotations were performed independently by five embryologists with a minimum of five years of experience, and the final class assignments were determined through majority voting to ensure label quality and consistency.

Together, these datasets offer a resource for developing and evaluating machine learning models for sperm detection, morphology analysis, and chromatin integrity assessment. By providing both raw and expertly labeled data, this collection supports reproducibility, comparative benchmarking, and future advancements in automated fertility diagnostics.

Methods

This section outlines the procedures used for sample collection, dataset construction, and expert annotation. The dataset collection comprises three parts: a raw dataset of high-resolution sperm images and two labeled datasets derived from it. All samples were prepared using the sperm chromatin dispersion (SCD) method to visualize halo formation as an indicator of DNA fragmentation. The raw dataset was assembled to reflect the variability encountered in clinical sperm samples. From this base, we generated two labeled subsets (one for sperm detection and one for DNA fragmentation classification) using manual cropping and expert annotation. The labeling process involved independent assessments by five trained embryologists, with majority agreement used to assign final class labels.

Sample collection and preparation. Sperm samples were obtained from patients undergoing intracytoplasmic sperm injection (ICSI) at a fertility center. Prior to collection, participants were instructed to observe a period of sexual abstinence lasting 3 to 5 days. Semen samples were collected via masturbation into sterile, non-toxic containers and allowed to liquefy at room temperature for approximately 30 minutes to ensure homogeneity. Once liquefied, the samples were processed for further analysis, including DNA fragmentation evaluation using the sperm chromatin dispersion (SCD) method.

The SCD test is a widely adopted approach for assessing sperm DNA fragmentation. It relies on the visualization of halo structures formed around sperm heads following controlled denaturation and lysis. Sperm with intact DNA typically exhibit large, well-defined halos, whereas fragmented DNA results in smaller or absent halos. This morphological distinction allows for semi-quantitative evaluation of chromatin integrity under bright-field microscopy.

Informed consent. The present study is an analytical diagnostic study conducted at the Infertility Center of Fatemeh Hospital, Hamadan, between January 2023 and July 2024. All patients were fully informed about

the study objectives and procedures before enrollment. Written informed consent was obtained from each participant for the use of their semen samples and anonymized data in research, including evaluation with artificial intelligence–based diagnostic methods. Participation was entirely voluntary, and patients were assured of their right to withdraw at any stage without any impact on their treatment. Detailed explanations of the study purpose, procedures, and potential risks and benefits were provided both verbally and in writing, and participants had the opportunity to ask questions before signing the consent forms.

Ethics approval. Ethical approval for the conduct of this study, including the collection and use of human semen samples and related clinical data, was obtained from the Ethics Committee of Tarbiat Modares University (approval code: IR.MODARES.REC.1401.107). All procedures were carried out in accordance with the ethical standards of the institutional research committee and with the principles of the Declaration of Helsinki. Written informed consent was obtained from all participants for sample collection, data sharing, and publication of anonymized microscopy images.

Evaluation of DNA fragmentation. DNA fragmentation was evaluated using the Halo Sperm Kit (Idehvarzan Farda, Tehran, Iran), which implements the sperm chromatin dispersion (SCD) method. Initially, agarose gel was melted at 90 °C, then cooled and equilibrated to 37 °C before being mixed with the semen sample. The resulting mixture was smeared onto pre-cleaned slides, covered with coverslips, and chilled at 4 °C for 5 minutes to immobilize sperm cells within the micro-gel matrix.

After removing the coverslip, slides were sequentially treated with an acid denaturation solution (7 minutes) and a lysis buffer (15 minutes), then rinsed with distilled water for 5 minutes. Dehydration was performed using a graded alcohol series (70%, 90%, and 100%, each for 2 minutes). Following air drying, slides were stained for 5 to 7 minutes to enhance contrast. For each sample, a minimum of 200 spermatozoa were examined under bright-field microscopy, and the DNA fragmentation index (DFI%) was calculated as the percentage of sperm with fragmented DNA out of the total number assessed.

Raw image dataset. The raw dataset consists of 435 high-resolution stained images of human sperm samples, captured using bright-field microscopy at 1000 × magnification. All samples were prepared using the sperm chromatin dispersion (SCD) method to highlight DNA fragmentation levels via halo formation. Each image includes multiple sperm cells, along with non-sperm elements such as round cells and debris, reflecting the natural variability observed in clinical specimens.

The staining and imaging procedures were designed to preserve cell integrity and maximize visual contrast between chromatin halos and surrounding structures. The resulting dataset provides a diverse range of sperm morphologies and halo patterns, making it suitable for a wide variety of tasks, including image segmentation, unsupervised clustering, halo size quantification, and detection model training.

Labeled Dataset 1: Sperm vs Non-Sperm classification. To enable automated detection of sperm cells in bright-field microscopy images, we created a labeled dataset focused on binary classification: sperm versus non-sperm. This dataset was derived from the raw high-resolution images through manual cropping. A total of 650 image patches were extracted: 320 confirmed as sperm cells and 330 as non-sperm objects, including round cells, debris, and other background elements.

All samples were independently annotated by five embryologists, each with a minimum of five years of experience in clinical embryology. Final labels were assigned based on majority agreement among the annotators. The dataset is organized into two folders (one for each class) and is suitable for training and evaluating classification and detection models in preprocessing pipelines for fertility analysis.

Labeled Dataset 2: DNA fragmentation classification via halo size. The second labeled dataset focuses on classifying sperm cells based on chromatin integrity using halo size as a visual proxy. A total of 403 individual sperm cells were manually cropped from the raw images and assigned to one of four classes: large halo ($n = 103$), medium halo ($n = 107$), small halo ($n = 92$), or no halo ($n = 101$). These categories reflect differing levels of DNA fragmentation, as identified through the sperm chromatin dispersion (SCD) test. Representative examples of sperm cells from each halo size category are shown in Fig. 1.

In the SCD test, spermatozoa are classified into four categories according to the relative size of the dispersion halo surrounding the sperm head (Halo kit, Idehvarzan Farda Company, Tehran, Iran). Sperm with large halos (halo width similar to or larger than the minor diameter of the sperm head) are considered to have intact DNA. Medium halos (halo width smaller than the minor diameter of the head but still clearly visible) represent partially fragmented DNA. Small halos (narrow halos barely distinguishable from the sperm head) and absent halos (no dispersion halo visible) indicate high levels of DNA fragmentation. This classification system has been widely adopted in previous studies and is consistent with the manufacturer's instructions of the commercial SCD kit, allowing reproducibility and standardized assessment across laboratories²⁵.

Annotation was carried out independently by five embryologists, each with over five years of clinical experience. The final label for each image was determined using a majority voting scheme to reduce subjectivity and ensure consistency across the dataset. This dataset supports multiclass classification tasks aimed at automating DNA fragmentation analysis and developing decision-support tools for fertility diagnostics.

The study was approved by the Research Ethics Committee of Tarbiat Modares University (IR.MODARES.REC.1401.107). Written informed consent was obtained from all participants for the use of their semen samples and anonymized data in research, including data sharing and analysis with artificial-intelligence–based diagnostic methods. All procedures were conducted according to the institution's ethical standards and the principles of the Declaration of Helsinki (1964) and its later amendments. No identifying information was included in the dataset, and all data were fully anonymized before analysis and public release.

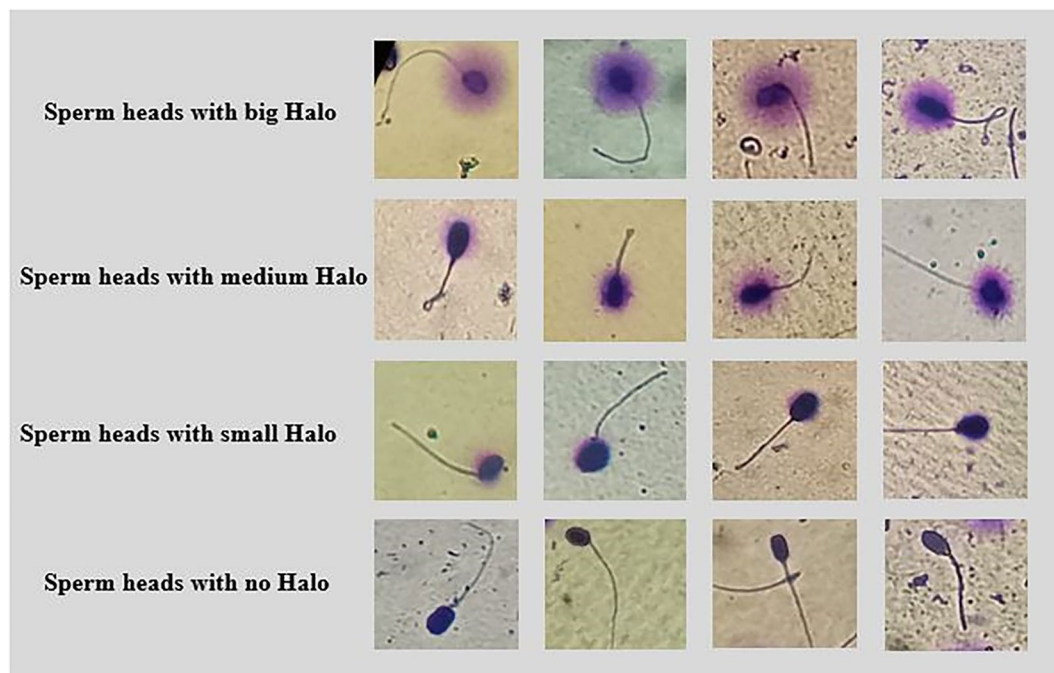


Fig. 1 Comparison of sperm with different halo sizes.

Data Records

The datasets introduced in this study including the raw image dataset and two labeled subsets²⁶ are publicly available through Figshare. The complete dataset collection is provided as a single ZIP archive. The archive includes:

- The raw dataset: full-field, high-resolution stained sperm images;
- A binary classification dataset: cropped image patches labeled as sperm or non-sperm;
- A multiclass classification dataset: individual sperm cells labeled by halo size (large, medium, small, or no halo).

The ZIP archive is organized into three main directories that correspond directly to the datasets described above. The Raw_Images/ directory contains full-field bright-field microscopy images, each including multiple sperm cells and background elements.

The Binary_Classification_Sperm_NonSperm directory corresponds to the binary classification dataset and contains cropped image patches organized into two subfolders (Sperm and NonSperm). The Halo_Classification directory corresponds to the multiclass halo dataset and contains cropped sperm images organized into five subfolders (LargeHalo, MediumHalo, SmallHalo, NoHalo, and NonSperm). All image files follow a standardized numerical naming convention (e.g., 001.png, 002.png, ...) without embedded identifiers.

The full dataset can be accessed and downloaded at the following link: <https://doi.org/10.6084/m9.figshare.30120811>.

A detailed README file inside the archive outlines the folder structure, labeling protocols, usage recommendations, and citation format. Any future updates or extended versions will also be shared via the same page.

Technical Validation

To ensure that the dataset could support robust analysis, we verified that halo size categories (large, medium, small, none) were consistently identified during annotation. Cropped sperm cell images were checked to confirm that the entire head and halo were preserved, while non-sperm entities such as debris and round cells were intentionally included in the binary classification subset to ensure realistic representation. Images also captured both dense and sparse fields of view, reflecting natural variation in sperm concentration across samples. These steps were designed to preserve the clinical heterogeneity of sperm morphology and fragmentation while minimizing annotation errors.

Cropped sperm cell images were checked to confirm that the entire head and halo were preserved, while non-sperm entities such as debris and round cells were intentionally included to ensure realistic representation. Examples of annotated images highlighting sperm (blue circles) and non-sperm objects (red circles) are shown in Fig. 2. These visual checks were performed to ensure accurate labeling and separation of relevant biological structures.

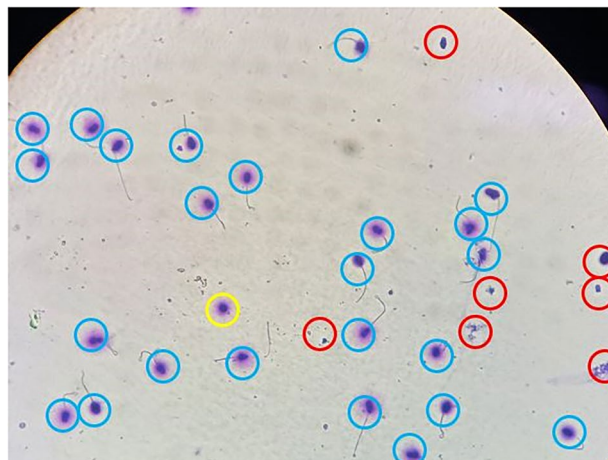


Fig. 2 Sample of Segmented Cells. Red circles are related to non-sperm cells and blue circles are related to sperm cells. The cell highlighted in the yellow circle seems to be a sperm head. However, its tail cannot be seen.

Usage Notes

The dataset structure supports a range of practical use cases, including sperm detection, segmentation, classification, and DNA fragmentation assessment. Researchers may preprocess images by resizing, cropping, or applying segmentation to isolate individual cells before training machine learning models. The availability of both raw and labeled data enables flexible experimental design, allowing users to benchmark methods for supervised learning, explore unsupervised clustering of sperm morphology, or validate automated counting pipelines for sperm concentration analysis.

Data availability

The dataset supporting this work is available at Figshare: <https://doi.org/10.6084/m9.figshare.30120811>.

Code availability

No custom code was generated or used during the current study.

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Author contributions

Hanan Saadat conceptualized the study, designed the methodology, and wrote the manuscript. Hossein Torkashvand and Mahdi-Reza Borna, collected and processed the dataset contributed to data analysis, manuscript editing, and discussions on dataset applications. All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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